

be set to position 4 as outlined. Power can then be applied to the circuit.

The outputs Q12 and Q13 of the CD4060BE are used to drive relays RL1 and RL2, respectively. When RL1 energises, it switches on the first AC load for approximately 17 minutes (1024 seconds). It then de-energises and switches off the AC for the next 17 minutes. Similarly, when RL2 energises, it switches on the second AC load for around 2048 seconds (34 minutes), and then de-energises and switches it off for the next 34 minutes.

This cycle continues indefinitely. The relays activate and deactivate according to the timing intervals selected via the rotary switch, as described in the table.

Construction and testing

An actual-size, single-sided PCB layout for the AC power saver is shown in Fig. 3, with the component layout in Fig. 4. Once the components have been assembled on the PCB, the circuit should be enclosed in a suitable cabinet. Ensure that appropriate holes are provided at the rear for mounting and ventilation. The rotary switch should be fixed to the front side of the cabinet and connected to the PCB using external wiring.

Alternatively, the circuit can be assembled on a general-purpose PCB. Once assembly is complete, the circuit should be placed inside a cabinet. A proper enclosure ensures protection from dust, moisture, and accidental damage, while also contributing to the reliable operation of the system.

Connections should be double-checked before powering the system. A 12V power supply should be connected to JACK1 via a 12V adaptor to operate the circuit. Blinking of LED1 indicates that the circuit is ready for use.

Installation

Installation of the air-conditioner (AC) power saver is straightforward. Begin with the assembled PCB. Connect flexible wires from the PCB to switch S1, and mount the switch at the front of the PCB. Both AC units and the 230V AC supply should then be connected to the respective two-pin connectors, CON1 through CON4. All connections must be secure before testing and completing the installation. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-make-your-own-air-conditioner-ac-power-saver>

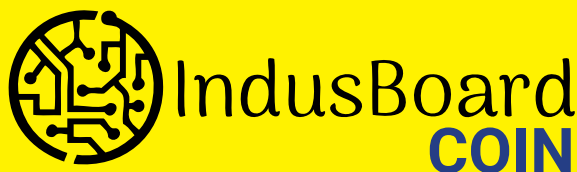
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